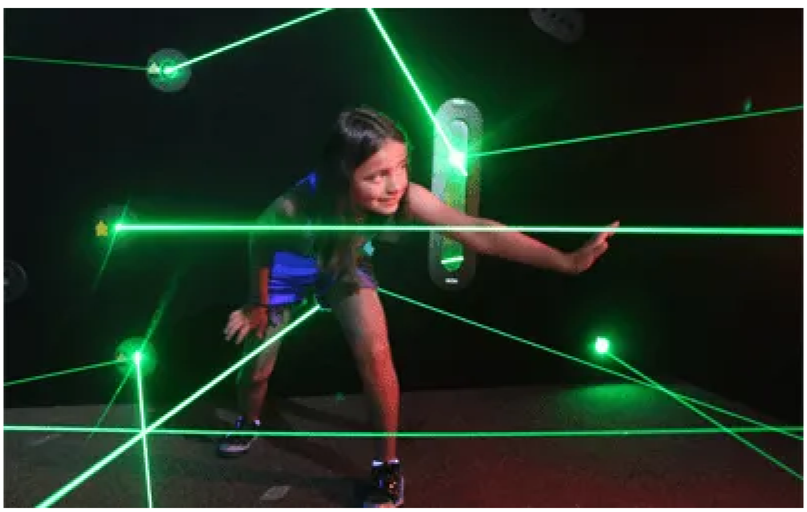
Laser Maze



Concept:

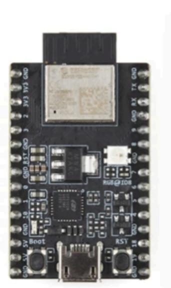
Kids can assemble the components:

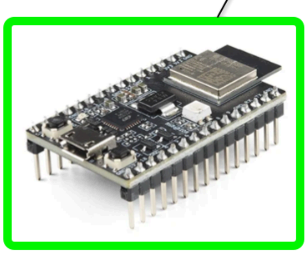
- **Goal**: no soldering
- **Easy concepts**
    - Send laser light
    - Reflect laser light
        - Multiple times
    - Sense laser light

        - Close circuit
    - Sound alarm
- **Fun**
    - Different layouts
    - Exercise
    - Pretend you’re the pink panther
    - Security for your secret hiding place
    - Keep your door open yet secure
- **Learnings**
    - Completing a circuit
    - Working with mirrors and reflections
    - What is a laser diode
    - What is a photo diode
    - What is a circuit
        - Electrical circuit
        - Light circuit (with mirrors)
- **Different Designs**
    - *Simple*: One mirror for simple door security
    - *Complex*: Perimeter with more mirrors
- ***Maybe Next Class***
    - *Add more components*
    - *Get a text message when the beam is obstructed!*

Soldering or No Soldering:

- ESP32 is the microcontroller we’re using





no soldering needed

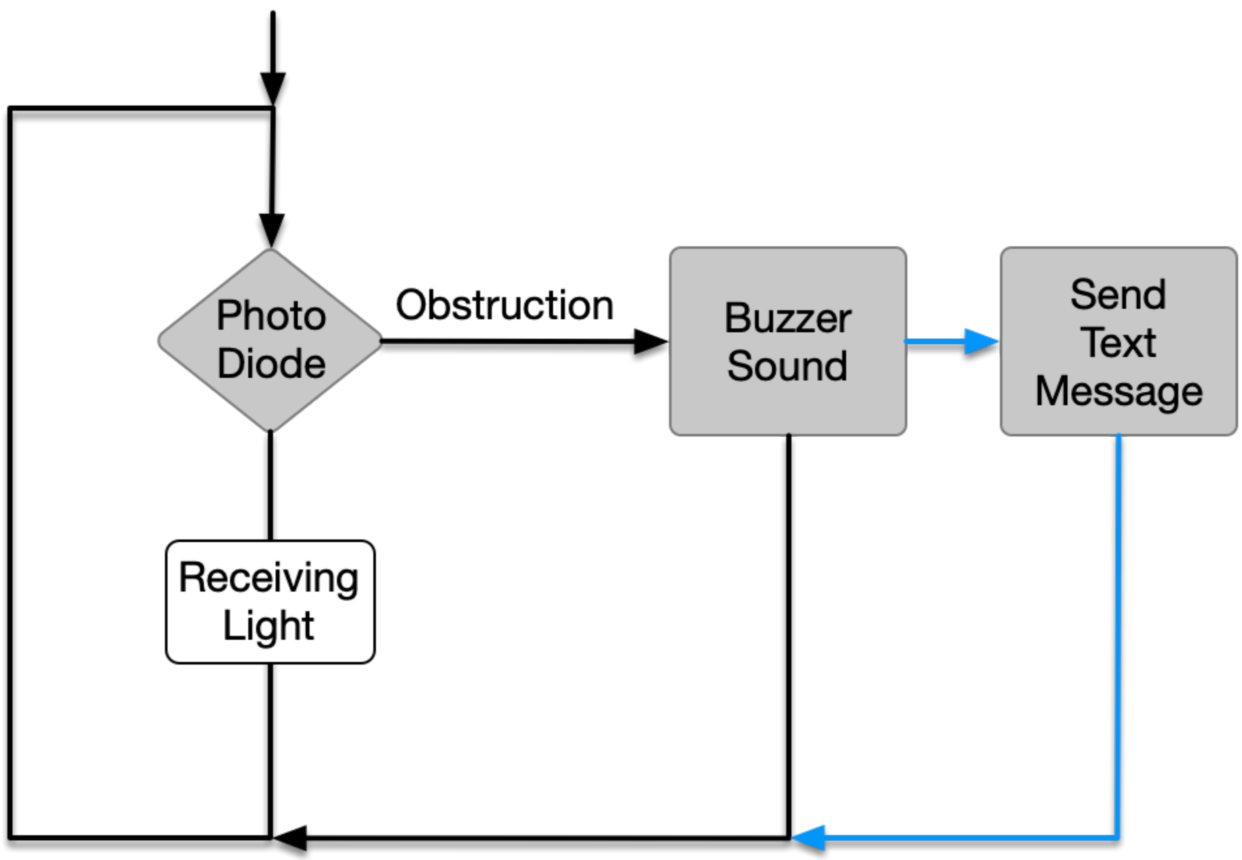
soldering needed

- To avoid soldering we use a breadboard, which takes up more space

Diagram:

A simple loop:

- Check if laser beam is broken
  - If broken, sound alarm
  - If not, continue



Code:

Arduino IDE (C++):

- [Laser-maze.ino](#) sketch (initially from AI)
- I asked [Google](#):
  - *how to hook up a laser diode, a photo diode, a 3.7 lipo, a buzzer and an esp32 mini show code*
  - It responded with the code below
    - Not tested
    - No hardware
    - But a plan !!!!!

C/C++

```cpp
#define LASER_PIN 13
#define PHOTO_PIN 34
#define BUZZER_PIN 12

// Threshold to detect if laser light is hitting photodiode
// (Adjust this value based on ambient light in your room)
const int THRESHOLD = 1000;

void setup() {
  Serial.begin(115200);
  pinMode(LASER_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(PHOTO_PIN, INPUT);
}

void loop() {
  digitalWrite(LASER_PIN, HIGH); // Turn on the laser
  delay(50); // Small delay to let the sensor read

  int sensorValue = analogRead(PHOTO_PIN);
  Serial.print("Photodiode Value: ");
  Serial.println(sensorValue);

  // If the light is blocked, the sensor value drops
  if (sensorValue < THRESHOLD) {
    digitalWrite(BUZZER_PIN, HIGH); // Sound the buzzer
    Serial.println("Laser Beam Broken! ALARM!");
  } else {
    digitalWrite(BUZZER_PIN, LOW);  // Turn off the buzzer
  }

  delay(200); // Wait between readings
}
```

Cost:

under $50 from Amazon

- Laser
- Mirrors
- Double stick tape
- Photo detector
- Battery
- Speaker
- Spray fog
- Wires
- Miscellaneous

Parts List:

- Laser
  - $10 for a good laser
- Batteries
  - $5
- Speaker
  - $5
- Mirrors
  - $6
- Photo Diode
  - $7
- Double Stick Tape
  - $10
- Fog Spray
  - $20
- Resistors
  - $10
- Wires
  - $6
- Miscellaneous

Option: Repurposing Existing Kits

There are kits available for around $50

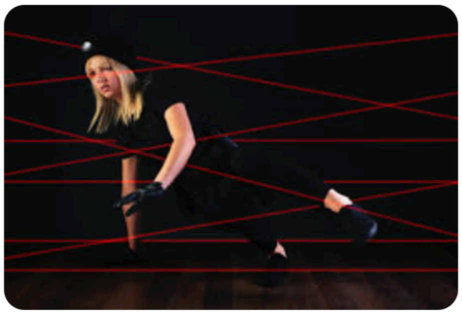
- Existing Laser Security Kits for Kids

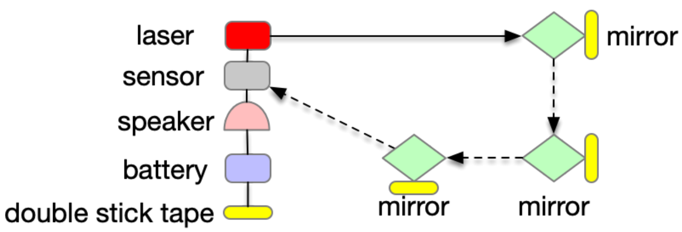
Shipping

- Everything can be ordered on Amazon and shipped directly to a third party (perfect)

Original sketch

Laser Maze





How to see the laser

[Diffusion Mist]() or [Fog in a Can]().

Amre Amer

Notes:

If you like, You could put up two more simple projects. Like basic science fair like projects for middle school and high school kids with **sensors** detecting a b c like **fire** or pollution etc one in climate tech or projects like smart **plant** watering system or **obstacle** avoiding robot or drawing robot or **clap active light** or **traffic light** simulator or delivery robot. We have good demand in summer months.

Please let me know which hardware to purchase you will need a copy of the hardware I assume.